

Research interests

Experienced in system modeling and big data analytics with a strong Physics and Mathematics foundation. Five years of proven expertise in clinical application systems implementation and optimization. Interdisciplinary problem solver passionate with expertise in image post-processing, data analysis, and statistical modeling across multiple programming languages. Proven ability to integrate diverse techniques and collaborate effectively in teams while adapting to new challenges and technologies.

Topics: Big data processing and analysis, deep learning, advanced modeling, breast, radiotherapy, probabilistic segmentation, fluid dynamics, multi wavelength imaging.

Research experience

Dic. 2021– **Senior AI Researcher at San Raffaele, Milano, Italy**

- present
- *Projects:* Prediction on toxicity/outcome of patient treatment through the development of automatic procedures (e.g. image anonymization, radiomic/densitometric/dosimetric feature extraction) for e.g., CT, MRI images. ML models (e.g., lightGBM) on large datasets (e.g., NET, breast, bowel, anesthesia) for classification, regression and to predict survival, with explainability analysis using SHAP. Bounding boxes and auto-segmentation of CTV breast with DL (e.g., nnUNET, SAM2). Inter and Intra center probability CTV map prediction temporally validated. Synthetic CT generation from MRI images using DL (cycleGAN, diffusion models, transformers) for SBRT.
 - Mentoring of 1 bachelor, 2 master, and 2 medical physics specialization thesis on DL segmentation models comparison, probabilistic map inter-center validation, cardiac toxicity prediction via DL models comparison.
 - *Collaborator to grants:*
 - PerPlanRT - Personalized planning in radiotherapy through integrative modeling of local dose effect and adapted sub organs constraints (founded by EU, ERAPERMED2020-110),
 - MIKAPOCO - Improving the quality of radiotherapy by multi-institution knowledge-based planning optimization models (founded by AIRC-IG23150),
 - Reducing toxicity after breast cancer Radiotherapy through predictive models and care personalization (founded by AIRC 2025),
 - TETRIS - Risk assesment tools for severe side effects after breast radiotherapy: radiation safety through biological extended models and digital twins (founded by EU),
 - AI-based segmentation of the CTV in breast radiotherapy: exploration of scalable models, including intra- and inter-center probability maps (founded by CCM 2024),
 - Integration of multiomics markers for invasive IPMNs identification through the set-up of the INvasive Cyst Biomarkers detection (INCITE) consortium (founded by PNRR)
 - *Ongoing collaborations:* C. Fiorino (supervisor), Prof. R. Jeraj (Ljubljana, Madison), L. Placidi (Gemelli), T. Rancati (INT), Prof. S. Carrazza (Unimi), Prof. M. Potenza (Unimi)
 - *Scientific visitor:* Ljubljana (May 2023), collaborators: Prof. R. Jeraj, L. Rivetti.
 - *Maternity leave:* 1 September 2024 - 1 March 2025

June 2019 – **Secondment founded by the RISE-2018 project DUSTBUSTERS, Melbourne,**
August 2019 Australia

Collaboration with Prof. D. Price, Dr. C. Pinte and V. Christiaens using VIP, PHANTOM and MCFOST.

May 2017 – **1 year ESO studentship, Garching, Germany**

June 2018 SPHERE data reduction under the supervision of Dr. M. van den Ancker

Jan. – Apr. **Scientific visitor at Leiden University, Leiden, The Netherlands**

2017 Collaboration with A. Miotello and S. Facchini on the modelling using DALI of a disk observed by ALMA.

- July 2016 **Scientific visitor at ESA/ESTEC, Leiden, The Netherlands**
Collaboration with Dr. C.F. Manara
- 05-06/16 & **Research scholar, Space Telescope Science Institute (STScI), Baltimore, USA**
09/15–02/16 Collaboration with the team of "The Orion Nebula Cluster as a Paradigm of Star Formation" HST project, under the supervision of Dr. M. Robberto (STScI & JHU)
- Aug.–Sept. **Internship, Leibniz Graduate School for "Quantitative Spectroscopy in Astrophysics",**
2014 **Potsdam, Germany**
Project: "Echelle Spectroscopy with SOPHIE of HD123351". Advisor: Dr. M. Steffen

Other work experience

- Febr. – Dic. **Permanent Data Scientist and Software Developer** at ATG Artificial Intelligence
2021 Srl, Erba. Study of the consistency of temporal correlations between adjacent frames; Predictions on large biomedical datasets using Decision Trees, Random Forest, Neural Network, Logistic Regression. Large time series dataset prediction using DL (e.g., LSTM).
- Sept.–Nov. **Freelance work as Data Scientist** at AISent Srl, Dalmine. Application of Computer Vision on healthcare solutions through CNN (i.e., Chest radiography illnesses identification, Brain Tumor Segmentation); tracking multi-camera aimed at people re-identification in a crowded scenario.
- Febr.–June **Associate Editor** of two scientific outreach exhibitions: "Explorers of the Solar
2014–2018 System" and "Exoplanets" presented at an International Fair: a cultural event visited by more than 800,000 people each year.
- 2010–2017 **Private lessons, I gave private lessons of Physics and Maths for high school students**
- Aug.2010– **Hostess/VIP Personal Assistant, the International Meeting of Rimini, Guests'**
2013 agenda management, trip booking, interview scheduling, tour guide
- June 2007 **Administrative support, clothes company Terranova**

Education

- Oct.2016– **PhD in Physics, Astrophysics and Applied Physics, University of Milan, Via**
present **Giovanni Celoria, 16, 20133 Milano, Italy**
Topic: "The role of (sub-)stellar companions on the dynamical evolution of protoplanetary disks". Advisor: Prof. Giuseppe Lodato (Univ. Milano). Dissertation expected in **July 2020**. Exams: Computational Physics, Atmospheric Dynamics, Advanced topics in Astrophysics and plasma physics.
- Sep.2013– **MSc in Physics [Dottore Magistrale in Fisica] Specialty: Astrophysics, Uni-**
April.2016 **versity of Milan, Italy. Final grade: Final grade: 110/110**
Thesis: "The sub-stellar mass function in the Orion Cluster". Advisors: Prof G. Lodato (Univ. Milano), M. Robberto (Space Telescope Science Institute (STScI) & Johns Hopkins Univ. (JHU))
- Sep.2009– **BSc Physics [Dottore in Fisica], University of Milan, Italy. Final grade: 101/110**
April.2013 Thesis: "Study of the kinetics of DNA denaturation by fluorescence techniques"
- 2004–2009 **High School Diploma in Classical Studies, Liceo classico "Dante Alighieri",**
Rimini, Italy

Further education

- 23-24 Oct. **Awarded ESTRO Physics Workshop 2025: Towards the Use of Probabilities for**
2025 **Treatment Prescription, Optimisation, and Evaluation of Radiation Treatment Plans.**
- Oct. 2021 **Courses finalized: Machine Learning Course and Deep Learning specialization on**
Coursera.
- May. 2019 **Exams: Psychology, Cultural anthropology**
- Febr. 2018 **Awarded ESO Observing School at La Silla, Santiago, Chile**

- August 2017 **St Andrews Monte Carlo Summer School**, St Andrews
- Sep.- **Intensive English Language Course**, *Emerald Cultural Institute*, **Dublin**, Ireland,
- Oct.2013 Pre-Intermediate B English CEFR A2 level.
- 2007-2008 **Astronomy classes**, at *University of Bologna*, Italy

Publications

First/co-first/corresponding(*) author publications

- 2026 "Probabilistic breast CTV maps derived from multi-institutional (MICAPOCO) contouring variability" **Ubeira-Gabellini***, **MG**; Bordigoni B.; Palazzo, G; (...); Fiorino, C, **in prep.**
- 2026 "Deep ensemble optimized models for probabilistic CTV breast segmentation" C. Riani, **Ubeira-Gabellini* (co-first)**, **MG**; (...); Fiorino, C. **Frontiers in Artificial Intelligence, section Medicine and Public Health**, publ. 9:1777653; DOI: 10.3389/frai.2026.1777653
- 2026 "Probabilistic CTV breast maps incorporating Inter-Observer Variability: Training on large datasets with clinical validation" **Ubeira-Gabellini***, **MG**; Riani C; Palazzo, G; (...); Fiorino, C **International Journal of Radiation Oncology**, **subm.**
- April 2025 "Development and external multicentric validation of a deep learning-based clinical target volume segmentation model for whole-breast radiotherapy" **Ubeira-Gabellini***, **MG**; Palazzo, G; (...); Fiorino, C **PHYSICS & IMAGING IN RADIATION ONCOLOGY**, **publ.**, DOI: 10.1016/j.phro.2025.100749
- March 2024 "Comparing Performances of Predictive Models of Toxicity after Radiotherapy for Breast Cancer Using Different Machine Learning Approaches" **Ubeira-Gabellini***, **MG**; Mori, M; (...); Fiorino, C. **CANCERS**, **publ.**, DOI: 10.3390/cancers16050934
- Jan. 2020 "Discovery of a low-mass companion embedded in the disk of the young massive star MWC 297", **M. G. Ubeira Gabellini***, V. Christiaens, G. Lodato, D. Fedele, M. van den Ancker, C.F. Manara, D. Price, 2020 **ApJL**, **publ.** , 890, 1, L8.
- April 2019 "A dust and gas cavity in the disc around CQ Tau revealed by ALMA", **M. G. Ubeira Gabellini***, A. Miotello, S. Facchini, E. Ragusa, G. Lodato , L. Testi, et al. 2019, **MNRAS paper**, **publ.**, 486, 4638-4654.

Last/Co-last author publications

- 2026 "ResMIP: A python-based Research library for Medical Image Processing (MIP): an example for radiomical application", G. Palazzo, A. Belardo, (...), **MG. Ubeira-Gabellini** (co-last), C. Fiorino, **in prep.**
- 2026 "Predicting cardiotoxicity using cardiac substructures and calcification metrics: insights from the REQUITE dataset" Volonteri T.; Fois, M.; (...); **Ubeira-Gabellini MG** (co-last), Fiorino, C, **in prep.**

Co-author publications

- 2026 "AI-based metrics of contouring consistency in breast cancer patients predict radiotherapy outcome", G. Palazzo, **MG. Ubeira-Gabellini**, A. del Vecchio, A. Fodor, C. Fiorino, **subm.**
- 2026 "Vetr-AI: an End-to-End Python toolbox for Whole Slide Image analysis in Digital Pathology", G. Palazzo, G. Ricciardi, **MG Ubeira-Gabellini**, F. Scarfò, C. Doglioni, M. Ponzoni, C. Fiorino, **in prep.**
- 2026 "Radiation-induced Lymphopenia is associated with 2-year patient-reported Intestinal Toxicity in Patients Treated with Radiotherapy including Pelvic Node Irradiation for Prostate Cancer" C. Cozzarini, G. Ricciardi, ...(including **M. G. Ubeira-Gabellini**)..., T. Rancati, C. Fiorino, **Radiotherapy and Oncology**, **subm.**

- 2026 "Association of Cardiac and Pulmonary CT Imaging Features with Respiratory Side Effects after Whole-Breast Radiotherapy" M. Fois, A. Belardo, A. Fodor, ...(including **M. G. Ubeira-Gabellini**)...; N. G. Di Muzio, T. Rancati, C. Fiorino **Cancers**, publ., 18(11), 1727; <https://doi.org/10.3390/cancers18111727>
- 2026 "Pancreatic Adenocarcinoma Nutritional-Clinical INDEX (PANCIN): a novel prognostic tool in advanced pancreatic cancer" Carconi C., Capurso G., Abati M., Vincenzi M.M., Burini A., Cardellini S., Pavarini M., **Ubeira-Gabellini M.G.**, (...), Fiorino C., Falconi M., Reni M., Orsi G., **Digestive and Liver Disease**, publ., DOI: <https://doi.org/10.1016/j.dld.2026.02.015>
- 2026 "Limited impact of breast densitometry in predicting toxicities after moderately hypo-fractionated whole-breast Radiotherapy" A Belardo, M Mori, A Fodor, M M Vincenzi, P Mangili, G Palazzo, **MG Ubeira Gabellini**, (...) C Fiorino, **Physica Medica**, **subm.**
- 2026 "Performances of AI-based automatic delineation of CTV and Organs-at-Risk for breast radiotherapy planning: including inter-observer variability in the evaluation process", F. S. Maddaloni, S. Broggi, A. Fodor, G. Palazzo, **M.G. Ubeira-Gabellini**, (...), C. Fiorino, **Physica medica**, **subm.**
- March 2026 "Predictors of acute lymphopenia after Radiotherapy for prostate cancer including pelvic node irradiation: Results of a real-world prospective multi-centric study" V. Vavassori, M. Pavarini, L. Alborghetti (... including **MG Ubeira Gabellini**) C. Cozzarini **ESMO Real World Data and Digital Oncology**, doi: 10.1016/j.esmorw.2025.100680, **publ.**
- March 2025 "Training and temporally validating an NTCP model of acute toxicity after whole breast radiotherapy, including the impact of advanced delivery techniques" Vincenzi, MM; Cicchetti, A; R. Castriconi; P. Mangili; **MG Ubeira Gabellini**(...); Fiorino, C **RADIOTHERAPY AND ONCOLOGY**, **publ.**
- June 2024 "Pelvic bone marrow dose-volume predictors of late lymphopenia following pelvic lymph node radiation therapy for prostate cancer" Pavarini, M; Alborghetti, L; (... including **MG Ubeira Gabellini**); Fiorino, C **RADIOTHERAPY AND ONCOLOGY**, **publ.**
- May 2024 "Skin dose-volume predictors of moderate-severe late side effects after whole breast radiotherapy" Cicchetti, A; Mangili, P; Fodor A.; **MG Ubeira Gabellini**; (...) Fiorino, C **RADIOTHERAPY AND ONCOLOGY**, **publ.**
- Oct. 2023 "Real-world validation of Artificial Intelligence-based Computed Tomography auto-contouring for prostate cancer radiotherapy planning" Palazzo, G; Mangili, P; (... including **MG Ubeira Gabellini**); Fiorino, C. **PHYSICS & IMAGING IN RADIATION ONCOLOGY**, **publ.**
- Oct. 2023 "Selective sparing of bladder and rectum sub-regions in radiotherapy of prostate cancer combining knowledge-based automatic planning and multicriteria optimization" Alborghetti, L; Castriconi, R; (...); Fiorino, C **PHYSICS & IMAGING IN RADIATION ONCOLOGY**, **publ.**
- April 2023 "Deep learning-based segmentation of prostatic urethra on computed tomography scans for treatment planning" Cubero, L; García-Elcano, L; (...); Pascau, J **PHYSICS & IMAGING IN RADIATION ONCOLOGY**, **publ.**
- June 2023 "Prediction of the characteristics of aggressiveness of pancreatic neuroendocrine neoplasms (PanNENs) based on CT radiomic features" Mori, M; Palumbo, D; (...); De Cobelli, F **EUROPEAN RADIOLOGY**, 33(6), pp.4412-4421, **publ.**
- June 2023 "Dynamical detection of a companion driving a spiral arm in a protoplanetary disk", Xie, C; Ren, BB; (...); **Ubeira-Gabellini, MG ASTRONOMY & ASTROPHYSICS**, , **publ.**

- Aug. 2022 "External or internal companion exciting the spiral arms in CQ Tau? Hammond, I; Christiaens, V; Price, D.; **MG Ubeira-Gabellini** (...); Fedele, D **MNRAS**, 515(4), pp.6109-6121, **publ.**
- April 2021 "A faint companion around CrA-9: protoplanet or obscured binary?" Christiaens, V; **Ubeira-Gabellini, MG**; (...); Reggiani, M **MONTHLY NOTICES OF THE ROYAL ASTRONOMICAL SOCIETY**, **publ.**
- April 2021 "A highly non-Keplerian protoplanetary disc. Spiral structure in the gas disc of CQ Tau", L. Wölfer, S. Facchini, (...), **M.G. Ubeira Gabellini, A&A paper, publ.**
- Oct. 2020 "On the Cavity Size in Circumbinary Discs", K. Hirsh, D. J. Price, J-F Gonzalez, **M. G. Ubeira-Gabellini** Ragusa, E., **MNRAS, publ.**, 498, 2936-2947, DOI 10.1093/mnras/staa2536
- June 2020 "HST Survey of the Orion Nebula Cluster in the H₂O 1.4 μ m Absorption Band. I. A Census of Substellar and Planetary-mass Objects", M. Robberto, M. Gennaro, **M. G. Ubeira Gabellini**, et al., **ASTROPHYSICAL JOURNAL**, **publ.**
- Sept. 2019 "Investigating the nature of the extended structure around the Herbig star RCrA using integral field and high-resolution spectroscopy", E. Rigliaco, R. Gratton, D. Mesa, (...), E. Sissa, **M. Ubeira Gabellini**, et al. 2019, **A&A paper publ.**, 632, A18.
- Feb. 2019 "Exploring the RCrA environment with SPHERE: Discovery of a new stellar companion", D. Mesa, M. Bonnefoy, R. Gratton, Van Der Plas, V. D'Orazi, E. Sissa, A. Zurlo, E. Rigliaco, T. Schmidt, M. Langlois, A. Vigan, **M.G. Ubeira Gabellini** et al. 2019, **A&A paper, publ.**, 624, A4.
- March 2017 New evidence for the dynamical decay of a multiple system in the Orion Kleinmann-Low Nebula, K.L. Luhman, M. Robberto, J.C. Tan, M. Andersen, **M.G. Ubeira Gabellini**, et al., 2017, **ApJ paper, publ.**, 838, L3.

Proceedings

- 2019 "An Unbiased SPHERE-IFS Survey of Nearby Herbig Ae/Be Stars: Are All Group I Disks Transitional?", **M.G. Ubeira Gabellini**, M. E. van den Ancker, D. Fedele, G. Lodato and C.F. Manara, **Proceedings IAU Symposium No. 345**
- 2018 "Planet formation in the ALMA era", Lodato, G., Veronesi, B., **Ubeira Gabellini, M.**, Ragusa, E., Franchini, A., Dipierro, G., Toward a Science Campus in Milan: A Snapshot of Current Research at the Physics Department Aldo Pontremoli, **Springer International Publishing**.

Not peer-reviewed publications

- Feb. 2019 "A new Companion Candidate around the Herbig Star V921 Sco", **M. G. Ubeira Gabellini**, M. van den Ancker, D. Fedele, G. Lodato, C.F. Manara, **RNAAS**, 2019, 3,4: <https://iopscience.iop.org/article/10.3847/2515-5172/ab1659/meta>.

Conference contributions, seminars

- 16 May 2026 **Contributed talk as Proffered Paper at ESTRO 2026**: "Clinical validation of AI-based probabilistic CTV breast maps incorporating Inter-Observer Variability", Stockholm, Sweden
- May 2026 **Poster at ESTRO 2026**: "Optimizing deep ensemble models for probabilistic CTV breast segmentation", Stockholm, Sweden
- Nov. 2025 **Internal Seminar at HSR radiotherapy clinical cases**: "Probability ctv breast map: training and clinical internal validation of an in-house model", Milan, Italy
- Sept. 2025 **Poster at CIBB**: "DL-based CTV segmentation: models' optimization and temporal validation for breast radiotherapy", Milan, Italy

- May 2025 **Mini-Oral ESTRO 2025**: "Incorporation of CTV delineation uncertainty through deep learning-based probabilistic segmentation maps", Vienna, Austria
- May 2025 **Invited seminar at first kick-off project's meeting at ISS**: "Development and external validation of a dl-based ctv segmentation model on a large multicentric dataset in the case of whole-breast radiotherapy treatment", Rome, Italy
- Oct. 2025 **Poster AIFM 2025** : "Incorporation of CTV delineation uncertainty through deep learning-based probabilistic segmentation maps", Verona, Italy
- July 2024 **Contributed talk at ICCR 2024**: "Development and external validation of a DL-based CTV segmentation model on a large multicentric dataset in the case of whole-breast radiotherapy treatment", Lyon, France
- June 2024 **Contributed talk at AIRO**: "Development and external validation of a DL-based CTV segmentation model on a large multicentric dataset in the case of whole-breast radiotherapy treatment", Milan, Italy
- May 2024 **Digital Poster ESTRO 2024**: "AI-based CTV segmentation on a large cohort of patients: the case of breast Radiotherapy", Glasgow, United Kingdom
- April 2024 **Poster** at retreat HSR
- March 2024 **Poster** at HSR Medical Physics colloquia on CTV breast segmentation
- Sept. 2023 **Invited Seminar** at Ljubljana University group meeting on CTV breast segmentation
- July 2023 **Contributed talk at AIFM 2023**: "Comparing ML performances of late toxicity predictive models on a large cohort of patients treated with whole breast Radiotherapy", Florence, Italy
- May 2023 **Internal Seminar** at HSR Medical Physics colloquia: "Basic concepts of neural network and applications to medicine", Milan, Italy
- May 2023 **Invited seminar** at Ljubljana University group meeting: "Comparing performances of predictive models of toxicity after radiotherapy for breast cancer using different ML approaches", Ljubljana, Slovenia
- May 2023 **Poster presentation ESTRO 2023**: "Comparing ML performances of acute toxicity predictive models on a large cohort of breast cancer patients", Vienna, Austria
- Dec 2019 **Contributed Talk** at "Milan Xmas Workshop on Star Formation": "A young and embedded low-mass companion around MWC 297", Milan, Italy
- July 2019 **Contributed Talk** at "Great Barriers in planet formation": "The gas and dust disk around the CQ Tau protostar", Palm Cove, Australia
- July 2019 **Contributed Talk** at "Disc-ussion workshop": "SPHERE survey of 21 Herbig stars", Melbourne, Australia
- 2019-06-13 **Invited seminar** at group meeting of SPH group Monash University, Melbourne, Australia.
- 2019-02-04 **Contributed talk** at DUST BUSTERS KICK-OFF MEETING: "The structure of disks and their possible interaction with planets through observation of SPHERE and ALMA, Università degli studi di Milano, Italy.
- Oct 2018 **POSTER PRICE** at the conference "Take a Closer Look. The innermost region of protoplanetary discs and its connection to the origin of planets", Munich-Garching. Title: "An Unbiased SPHERE-IFS Survey of Nearby Herbig Ae/Be Stars: Are All Group I Disks Transitional?"
- March 2018 **Poster presentation** at "SPF2: Star and Planet Formation in the Southwest", Oracle, Arizona, USA. Title: "The gas and dust disk around the CQ Tau protostar"
- 11-15 Dic. 2017 **Poster presentation** at the Symposium Exoplanets and Planet Formation: "The gas and dust disk around the CQ Tau protostar", Shanghai, China

- 22 Nov. 2017 **Contributed talk** to Wine & Cheese seminar on 'A SPHERE-IFS Survey of Nearby Herbig Ae/Be Stars: Are All Group I Disks Transitional?', Garching, Germany.
- 8-12 Aug. 2016 Participation and contribute to the talk: "The HST/WFC3 census of the Substellar Population of the Orion Nebula Cluster" at the **conference "Star Clusters: from Infancy to Teenagehood"**, Heidelberg, Germany
- 19 July 2016 **Breakfast Seminar presentation: "The sub-stellar mass function in the Orion Cluster"**, ESTEC, the Leiden, The Netherlands
- Jan. 2016 **Poster presentation, 227th Meeting of the American Astronomical Society (AAS), Orlando, Florida**, Title: "Reconstructing the low-mass IMF of the Orion Nebula Cluster through HST photometry in the H2O band at 1.4micron"
- Aug. 2015 **Poster presentation, Honolulu, Hawaii**, Title: "Toward a Complete Census of the Low Mass IMF in the Orion Nebula Cluster"

Awards, telescope proposal accepted and studentships

- 2026 - present Senior post-doc under IPMN (founded by PNRR), HSR
- 2025 Senior post-doc under MIKAPOCO (founded by AIRC) UniSR
- 2021-2024 Post-doc under PerPlanRT (founded by EU), HSR
- 2018 PI of the SPHERE proposal accepted: "Confirming the nature of recently discovered low-mass companions to the five nearby Herbig stars", 10h
- 2017 PI of the SPHERE proposal accepted: "Confirming the nature of recently discovered low-mass companions to four nearby Herbig Ae/Be stars", 8h.
- 2017-2018 1 year ESO studentship
- 2017 co-PI of the ALMA proposal accepted: "Disk-planet interaction in a nearby young system". PI: Leonardo Testi.
- 2016 Università di Milano graduate research fellowship
- 2016 Scientific visitor at ESA/ESTEC
- 2015-2016 STScI Research scholar
- 2014 Summer Internship at Leibniz Graduate School, Potsdam

Conference organization and outreach

- July 2022 Outreach talk for young people: "How do planet form?" Milan, Italy
- April 2022 Outreach Talk: "Insights on planet formation" Milan, Italy
- April 2019 Member of the **LOC** at DUST BUSTERS KICK-OFF MEETING: "The structure of disks and their possible interaction with planets through observation of SPHERE and ALMA", Università degli studi di Milan, Italy
- Oct. 2018 Member of the LOC for "Take a Closer Look. The innermost region of protoplanetary discs and its connection to the origin of planets", Munich-Garching, Germany.
- Feb-June 2014-2018 **Associate Editor** of two scientific outreach exhibitions: "Explorers of the Solar System" and "Exoplanets" presented at an International Fair: a cultural event visited by more than 800,000 people each year.

Professional Skills and Expertise

Development of medical pipelines on github: pymaitre	Private libraries (for images): pydicom-link: Automatic retrieval of patient imaging studies from the PACS for a specific research project, with subsequent storage in Orthanc. pydicom-toolkit: Python library for DICOM folder organization in batch (dcm_folder) and to anonymize dicom files in parallel (dcm_anonymize). resmip: Python library for Medical Image Processing (MIP) es. dicom-to-nifti conversion. Probs-LAB: Package for probabilistic segmentation using DL models. VetrAI: Python library for processing, analyzing, and preparing Whole Slide Images (WSIs) in digital pathology. Training of YOLO/Sam for tissue segmentation. Private libraries (for DBs): SQLHub: Code dealing with the interaction between analysis in Python and databases stored in SQL servers. medicalAI: database pre-processing: cleaning and features selection (e.g. bootstrap or backward feature selection); simple AI models for training and prediction; automatic saving plots and and model output storage (via excel/SQL; e.g., ROC, PR, calibration map, SHAP). medicalDL: more advanced model's implementation (es. lightGBM) for training and predictions.
Clinical skills	Radiotherapy, Radiology, Imaging
Astronomical skills	Infrared Photometry (with HST); Spectroscopy (comparison between observed and synthetic spectra); data reduction of images coming from Extreme adaptive optics systems and coronagraphic facility (SPHERE) with VIP; hydrodinamical and radiative transfer numerical simulations with PHANTOM and MCFOST.
Op. Systems	OSX, Linux, Windows
Text editing	LaTex, MS Office
Scientific packages	Python, C++ (also with Root), IDL, SAOImage DS9, Mathematica, Origin, TopCat, Glue, Gaia skycat

Languages

Italian	Mothertongue
English	Fluent, SLAM (certificate: C1 level)
Spanish	Basic

Interests and Other activities

2009–2016	Volunteer , “ <i>Cesano Boscone</i> ”, institute for disabled people in a suburb of Milan; I gave private lessons at the center for study support “Portofranco” in Milan
2009–2015	University press club , <i>selection and comparison of articles on world events from major newspapers</i> , Editor of the University Student’s Newspaper
Dec. 2009–2014	Fundraiser , <i>AVSI (The Association of Volunteers in International Service)</i>